

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (CE/IT)

Dec 2012

IT405 Advanced Computing

Date: 08.12.2012, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Google App Engine is an example of Software as a Service (SAAS). State true or false. Justify your answer with proper example. [05]
- (b) Explain how cloud computing application differs from normal web application. [02]
- Q - 2 (a) What is hypervisor? Compare the Xen and Oracle Virtual box virtualization soft wares. [04]
- (b) List and explain the service models of cloud computing. [07]
- (c) What are the types of virtualization? Explain any one of them. [03]

OR

- Q - 2 (a) Draw the architecture of Xen hypervisor. Why Domain 0 and Domain U both needed in Xen architecture? [04]
- (b) What are the deployment models of the cloud computing? Write the differences between the deployment models. [07]
- (c) Write the advantages of Virtualization. [03]
- Q - 3 (a) Write down the steps to Setup compute cluster using MPICH library in Ubuntu. [06]
- (b) What is openMP? Explain the need of pragma in it. [04]
- (c) List the controller of eucalyptus. Explain any one in brief. [04]

OR

- Q - 3 (a) What are job tracker and task tracker in Hadoop? What is the purpose of combiner function? List the Areas where Hadoop Distributed File System (HDFS) is not a good fit. [06]
- (b) State the differences between openMP and MPI? [04]
- (c) Explain the send and receive primitives of MPI. [04]

SECTION – II

- Q - 4 (a) Differentiate process migration and computation migration. [03]
(b) Does Open MPI support checkpoint and restart of parallel jobs? Justify your answer. [04]
- Q - 5 (a) Give the difference between non-dedicated clusters and dedicated clusters. Explain any one non-dedicated cluster. [06]
(b) Comments on the advantages of using a virtual grid platform against traditional HPC systems. [05]
(c) What is the difference between Grid Computing and distributed computing? [03]

OR

- Q - 5 (a) What is high-performance cluster? Explain in terms of network, application and computing. [06]
(b) Explain Open grid service architecture. [05]
(c) How decentralized non-coordinated scheduling work in grid computing environment? [03]
- Q - 6 Write a short note on any TWO. [14]
a. How policy services provide a set of rules to administer, manage and control access to Grid resources? Explain with architecture.
b. Explain the architecture of ChinaGrid Service Supporting Platform.
c. List out parallel computer architectures. Explain any two them with their advantages and disadvantages.
